

Module 4 COMSOL FEM Implementation Note

Click-by-Click Beginner Tutorial for RadiationEffects_proj2

Parallel COMSOL implementation of two-dimensional transient thermal transport with radiation heating and reduced ballistic-to-diffusive source spreading

Purpose of this note. This document is a standalone, beginner-level COMSOL Multiphysics implementation guide for the finite element method (FEM) version of Module 4. The main project implementation remains MATLAB, but this note shows how to build a parallel COMSOL Multiphysics model for the same physical problem: transient thermal transport in silicon driven by radiation energy deposition, defect-related heat sources, and optional ballistic-to-diffusive source spreading. The goal is to make the procedure reproducible by someone who has never built a COMSOL Multiphysics model before. The instructions are written in the same spirit as official COMSOL Multiphysics tutorials: create the model, define parameters, build geometry, assign materials, set physics, mesh, solve, plot, verify, and export. The note intentionally includes many explicit menu paths, expression tables, verification worksheets, and common-error checks.

Project convention. This COMSOL Multiphysics model is a parallel implementation path only. It does not replace the MATLAB FEM solver. Use it as a check on the physics, boundary conditions, source normalization, mesh convergence, time-step sensitivity, and interpretation of the Module 4 heat equation.

Contents

Symbols and acronyms used in this document	7
1 Physical problem solved by Module 4	9
1.1 What Module 4 does in the project	9
1.2 Core heat equation	9
1.3 Fourier thermal diffusivity	10
1.4 What “ballistic-diffusive” means here	10
2 FEM interpretation before opening COMSOL	10

2.1	Weak form of the heat equation	10
2.2	Finite element matrix form	11
3	Recommended COMSOL model route	11
4	Route A: Heat Transfer in Solids, click-by-click	12
4.1	Start a new model	12
4.2	Save the model	12
5	Define global parameters	12
5.1	Open the parameter table	12
6	Build the two-dimensional silicon geometry	14
6.1	Create a rectangle	14
6.2	Name the geometry selection	14
7	Define source variables	14
7.1	Open component variables	14
8	Assign silicon material properties	15
8.1	Create a blank material	15
8.2	Enter material values	16
9	Set up Heat Transfer in Solids	16
9.1	Check the domain equation	16
9.2	Initial temperature	16
9.3	Add a volumetric heat source	17
10	Boundary conditions	17
10.1	Default thermal insulation	17
10.2	Verify the default insulation	17
10.3	Add fixed-temperature contacts, optional	17
11	Mesh creation	18

11.1	Create a controlled triangular mesh	18
11.2	Mesh physics explanation	19
11.3	Optional local refinement near the hotspot	19
12	Set the time-dependent study	19
12.1	Open Study 1	19
12.2	Solver tolerance	19
12.3	Compute	20
13	Postprocessing and plots	20
13.1	Temperature surface plot	20
13.2	Temperature rise plot	20
13.3	Line plot through the hotspot	21
14	Derived values and energy checks	21
14.1	Create an integration operator	21
14.2	Total excess thermal energy	21
14.3	Evaluate global quantities	22
15	Verification tutorial 1: uniform equilibrium	22
15.1	Purpose	22
15.2	Steps	22
15.3	Expected result	23
16	Verification tutorial 2: uniform source in insulated domain	23
16.1	Purpose	23
16.2	Steps	23
16.3	Expected result	23
17	Verification tutorial 3: Gaussian diffusion	24
17.1	Purpose	24
17.2	Create a Gaussian initial condition	24
17.3	Expected result	24

18 Verification tutorial 4: fixed-temperature boundaries	24
18.1 Purpose	25
18.2 Steps	25
18.3 Expected result	25
19 Route B: Coefficient Form PDE implementation	25
19.1 When to use this route	25
19.2 Create the PDE model	25
19.3 Coefficient form equation	26
19.4 Enter PDE coefficients	26
19.5 Boundary condition in PDE route	27
20 Adding imported radiation-deposition maps	27
20.1 Why import maps	27
20.2 Prepare a text file	27
20.3 Import interpolation function	27
20.4 Use imported map in source term	28
21 Adding defect-related thermal coupling	28
21.1 Temperature-dependent defect kinetics	28
21.2 Defect reaction heat source	28
22 Exporting data to MATLAB	29
22.1 Export a temperature field	29
22.2 Export centerline data	29
23 Mesh convergence workflow	29
23.1 Why convergence matters	29
23.2 Manual mesh convergence steps	29
24 Time-step sensitivity workflow	30
24.1 Why time-step checks matter	30
24.2 Steps	30

25 Common mistakes and fixes	30
26 Mapping COMSOL objects to MATLAB FEM objects	31
27 Recommended model-building order	32
28 Final trust checklist	32
29 Detailed COMSOL node tree expected after setup	33
30 Detailed expression-entry rules in COMSOL	33
30.1 Multiplication signs	34
30.2 Units inside brackets	34
30.3 Coordinate variables	34
30.4 Time variable	34
31 More detailed boundary-condition options	34
31.1 Thermal insulation	34
31.2 Fixed temperature	35
31.3 Prescribed heat flux	35
31.4 Convective cooling	36
32 How to create selections for boundaries	36
32.1 Left boundary selection	36
32.2 Right boundary selection	36
32.3 Source-region selection, optional	37
33 How to inspect the mesh	37
33.1 Plot mesh quality	37
33.2 Check element size near the source	37
33.3 Check mesh statistics	37
34 How to duplicate studies for verification cases	37
35 Parametric sweep for mesh refinement	38

35.1 Add the sweep	38
35.2 What to record	38
36 Parametric sweep for ballistic source width	38
37 Source normalization issue	39
38 Interpreting 2D units carefully	39
39 Adding temperature-dependent thermal conductivity	40
40 Adding a heat sink layer, optional model variant	40
41 Creating probes for time traces	41
41.1 Point probe at hotspot center	41
41.2 Domain probe for total thermal energy	41
42 Report generation inside COMSOL, optional	41
43 How to compare with MATLAB Module 4	42
43.1 Use the same inputs	42
43.2 Compare robust quantities first	42
43.3 Then compare field maps	42
44 Beginner navigation summary	42
45 Detailed verification worksheets	43
45.1 Worksheet A: baseline radiation pulse case	43
45.2 Worksheet B: local deposition only	44
45.3 Worksheet C: ballistic broadening sensitivity	44
45.4 Worksheet D: insulated energy accounting	45
46 Detailed solver settings for difficult transient runs	45
46.1 Open the solver configuration	45
46.2 Suggested time stepping	46

46.3 Linear solver choice	46
47 Detailed instructions for saving model variants	46
47.1 Use separate files for major variants	46
47.2 Use descriptive study names	46
48 Detailed imported-map workflow	47
48.1 Check coordinate convention before import	47
48.2 Convert deposited energy to heat source	47
48.3 Interpolation settings to check	47
49 Detailed postprocessing recipes	48
49.1 Plot temperature at selected times	48
49.2 Create an animation	48
49.3 Export plot images	48
50 Physical interpretation of common outcomes	48
50.1 High narrow peak	48
50.2 Broad low-amplitude hotspot	49
50.3 Boundary-dominated cooling	49
50.4 Nearly uniform temperature rise	49
51 What not to include in the first COMSOL model	49
52 Standalone reproduction checklist	50
53 Summary	50

Symbols and acronyms used in this document

Symbol	Meaning	Units
T	Temperature field solved in Module 4.	K

Symbol	Meaning	Units
ρ_m	Mass density of silicon or effective simulated material.	kg/m ³
c_p	Specific heat capacity at constant pressure.	J/(kg K)
k	Thermal conductivity.	W/(m K)
α_{th}	Thermal diffusivity, $\alpha_{th} = k/(\rho_m c_p)$.	m ² /s
Q_{tot}	Total volumetric heat source.	W/m ³
Q_{dep}	Local radiation-deposition heat source.	W/m ³
Q_{ball}	Reduced ballistic-to-diffusive heat source or broadened thermalization source.	W/m ³
Q_{defect}	Optional heat source or sink from defect reactions and annealing.	W/m ³
$Q_{carrier}$	Optional heat source from carrier recombination or Joule heating.	W/m ³
Ω	Two-dimensional silicon computational domain.	m ²
$\partial\Omega$	Boundary of the two-dimensional domain.	m
$\hat{\mathbf{n}}$	Outward unit normal vector on the boundary.	1
L_x, L_y	Width and height of the two-dimensional domain.	m
h	Representative mesh element size.	m
N_a	FEM shape function associated with local node a .	1
M_T	Heat-capacity mass matrix.	J/K per out-of-plane thickness
K_T	Thermal conductivity matrix.	W/K per out-of-plane thickness
F_T	FEM heat-source load vector.	W per out-of-plane thickness
Δt	Time-step size or output-time spacing.	s
η_{ball}	Fraction or scaling factor assigned to the broadened ballistic source.	1
σ_{dep}	Width of local deposition source.	m
σ_{ball}	Width of broadened ballistic source.	m

1 Physical problem solved by Module 4

1.1 What Module 4 does in the project

Module 4 connects radiation energy deposition and defect evolution to the local temperature field in silicon. A radiation event deposits energy. Some energy is thermalized locally, some can be represented as a broadened ballistic source before becoming diffusive heat, and some thermal response can feed back into defect evolution through temperature-dependent diffusion and annealing.

At the continuum level, Module 4 solves for the temperature field

$$T(x, y, t), \quad (1.1)$$

where x and y are the two spatial coordinates and t is time. The result is passed to other modules as a temperature map.

1.2 Core heat equation

The reduced FEM equation used in this tutorial is

$$\rho_m c_p \frac{\partial T}{\partial t} - \nabla \cdot (k \nabla T) = Q_{\text{tot}}(x, y, t), \quad (1.2)$$

where ρ_m is mass density, c_p is specific heat, k is thermal conductivity, and Q_{tot} is the total volumetric heat source.

For this project the total source can be decomposed as

$$Q_{\text{tot}} = Q_{\text{dep}} + Q_{\text{ball}} + Q_{\text{defect}} + Q_{\text{carrier}}, \quad (1.3)$$

where Q_{dep} is local radiation-deposition heating, Q_{ball} is an optional broadened ballistic-thermalization source, Q_{defect} is heat from defect reactions or annealing if included, and Q_{carrier} is optional carrier recombination or Joule-related heat if later coupled.

For the first COMSOL Multiphysics model we use

$$Q_{\text{tot}} = Q_{\text{dep}} + Q_{\text{ball}}, \quad (1.4)$$

then add the other terms as extensions.

1.3 Fourier thermal diffusivity

The thermal diffusivity is

$$\alpha = \frac{k}{\rho_m c_p}. \quad (1.5)$$

This scale controls how fast a temperature feature spreads. A localized hotspot of width σ_0 spreads roughly as

$$\sigma^2(t) \sim \sigma_0^2 + 4\alpha t \quad (1.6)$$

in two-dimensional diffusion. This relation is useful for verifying the simulation.

1.4 What “ballistic-diffusive” means here

A full microscopic ballistic phonon model would require solving a transport equation for phonon distributions. That is not what this first COMSOL Multiphysics implementation does. Instead, the COMSOL Multiphysics model keeps a continuum heat equation but allows the radiation source to be spatially or temporally broadened to represent energy carried away from the deposition site before local thermalization.

In practice, this means that the source term can be written as a sum of a narrow local deposition term and a broader source term:

$$Q_{\text{dep}}(x, y, t) = Q_0 f_{\text{narrow}}(x, y) g(t), \quad (1.7)$$

$$Q_{\text{ball}}(x, y, t) = \eta_{\text{ball}} Q_0 f_{\text{broad}}(x, y) g(t), \quad (1.8)$$

where f_{broad} has a larger spatial width than f_{narrow} . This is a reduced source-level representation of ballistic spreading.

2 FEM interpretation before opening COMSOL

2.1 Weak form of the heat equation

Multiply Eq. (1.2) by a test function w and integrate over the domain Ω :

$$\int_{\Omega} w \rho_m c_p \frac{\partial T}{\partial t} d\Omega - \int_{\Omega} w \nabla \cdot (k \nabla T) d\Omega = \int_{\Omega} w Q_{\text{tot}} d\Omega. \quad (2.1)$$

Integrating the diffusion term by parts gives

$$\int_{\Omega} w \rho_m c_p \frac{\partial T}{\partial t} d\Omega + \int_{\Omega} k \nabla w \cdot \nabla T d\Omega = \int_{\Omega} w Q_{\text{tot}} d\Omega + \int_{\partial\Omega} w k \nabla T \cdot \mathbf{n} d\Gamma. \quad (2.2)$$

For thermally insulating boundaries,

$$-k\nabla T \cdot \mathbf{n} = 0, \quad (2.3)$$

so the boundary term vanishes.

2.2 Finite element matrix form

Approximate the temperature in each element by shape functions N_a :

$$T_h(x, y, t) = \sum_a N_a(x, y) T_a(t), \quad (2.4)$$

where $T_a(t)$ is the nodal temperature. The semidiscrete FEM system becomes

$$M \frac{d\mathbf{T}}{dt} + K\mathbf{T} = \mathbf{F}, \quad (2.5)$$

where

$$M_{ab} = \int_{\Omega} \rho_m c_p N_a N_b d\Omega, \quad (2.6)$$

$$K_{ab} = \int_{\Omega} k \nabla N_a \cdot \nabla N_b d\Omega, \quad (2.7)$$

$$F_a = \int_{\Omega} N_a Q_{\text{tot}} d\Omega + \text{extboundaryheat} - \text{fluxterms}. \quad (2.8)$$

COMSOL Multiphysics constructs these FEM matrices internally. The point of the click-by-click procedure below is to make sure the physics entries correspond to Eq. (2.5).

3 Recommended COMSOL model route

There are two valid ways to implement Module 4 in COMSOL Multiphysics.

1. **Route A: Heat Transfer in Solids interface.** This is recommended if your COMSOL Multiphysics license includes the Heat Transfer Module. It uses the physical names *density*, *heat capacity*, *thermal conductivity*, *heat source*, and *thermal insulation*. This is easiest for beginners.
2. **Route B: Coefficient Form PDE interface.** This works with the Mathematics interface. It exposes the PDE coefficients directly and makes the weak-form/FEM connection clearer.

This note gives both routes. Start with Route A. Use Route B if you want a closer match to the MATLAB FEM equation assembly.

4 Route A: Heat Transfer in Solids, click-by-click

4.1 Start a new model

1. Open COMSOL Multiphysics.
2. In the opening window, click **Model Wizard**.
3. Under **Select Space Dimension**, click **2D**.
4. Click **Next**.
5. In **Select Physics**, expand **Heat Transfer**.
6. Expand **Heat Transfer in Solids**.
7. Select **Heat Transfer in Solids (ht)**.
8. Click **Add**.
9. Click **Study**.
10. Under **Preset Studies**, select **Time Dependent**.
11. Click **Done**.

You should now see the Model Builder tree with **Global Definitions**, **Component 1**, **Geometry 1**, **Materials**, **Heat Transfer in Solids**, **Mesh 1**, **Study 1**, and **Results**.

4.2 Save the model

1. Click **File** → **Save As**.
2. Navigate to a project working folder, for example `comsol/module4/`.
3. Save the file as `module4_thermal_2d_tutorial.mph`.

5 Define global parameters

5.1 Open the parameter table

1. In the Model Builder, expand **Global Definitions**.
2. Click **Parameters 1**. If it does not exist, right-click **Global Definitions** and choose **Parameters**.

3. In the Settings window, find the table with columns **Name**, **Expression**, **Value**, and **Description**.
4. Enter the parameters in Table 2 exactly as written.

Table 2: Global parameters to type in COMSOL Multiphysics.

Name	Expression	Description
Lx	20 [um]	Device length in the x -direction
Ly	10 [um]	Device height in the y -direction
T0	300 [K]	Initial temperature
Tleft	300 [K]	Optional left contact temperature
Tright	300 [K]	Optional right contact temperature
rhoSi	2330 [kg/m ³]	Silicon mass density
CpSi	700 [J/(kg*K)]	Silicon specific heat capacity
kSi	148 [W/(m*K)]	Silicon thermal conductivity near room temperature
alphaSi	kSi/(rhoSi*CpSi)	Thermal diffusivity
Q0	1e15 [W/m ³]	Peak volumetric source strength
etaBall	0.3	Fractional weight for broadened ballistic source
x0	0.5*Lx	Source center in x
y0	0.5*Ly	Source center in y
sigDep	0.6 [um]	Width of local deposition source
sigBall	2.5 [um]	Width of broadened ballistic source
tpulse	10 [ns]	Source pulse duration
tend	2 [us]	Final simulation time
dtout	10 [ns]	Output time spacing
hmax	0.4 [um]	Maximum mesh element size

Physics meaning. The product $\rho_m c_p$ controls how much energy is needed to change the temperature. The conductivity k controls how rapidly heat spreads spatially. The source strength Q_0 controls the rate at which energy is injected per unit volume.

6 Build the two-dimensional silicon geometry

6.1 Create a rectangle

1. In Model Builder, expand **Component 1**.
2. Click **Geometry 1**.
3. In the Settings window, set **Length unit** to **um**.
4. Right-click **Geometry 1**.
5. Choose **Rectangle**.
6. In the Rectangle settings, set **Width** to **Lx**.
7. Set **Height** to **Ly**.
8. Under **Position**, set **x** to 0 and **y** to 0.
9. Click **Build Selected**.

You should see one rectangular domain. This represents the 2D silicon cross-section.

6.2 Name the geometry selection

Selections are optional but useful.

1. Right-click **Definitions** under **Component 1**.
2. Choose **Selections** → **Explicit**.
3. Rename the selection **silicon_domain**.
4. In the Graphics window, click the rectangular domain.
5. Click the **+** button in the Settings window to add the domain to the selection.

7 Define source variables

7.1 Open component variables

1. In Model Builder, right-click **Component 1** → **Definitions**.
2. Choose **Variables**.

3. Rename it **Thermal source variables** if desired.
4. In the table, enter the variables in Table 3.

Table 3: Variables for local and ballistic source terms.

Name	Expression	Description
r2dep	$(x-x_0)^2 + (y-y_0)^2$	Squared distance from source center
fdep	$\exp(-r2dep/(2*\text{sigDep}^2))$	Narrow local deposition shape
fball	$\exp(-r2dep/(2*\text{sigBall}^2))$	Broadened ballistic shape
gtime	$\text{flc2hs}(tpulse-t, 0.1*tpulse)$	Smooth pulse active for $t < tpulse$
Qdep	$Q_0*fdep*gtime$	Local heat source
Qball	$\text{etaBall}*Q_0*fball*gtime$	Ballistic broadened source
Qtot	$Qdep+Qball$	Total source used by heat equation

Typing note. `flc2hs(arg,scale)` is a built-in smoothed Heaviside-type function in COMSOL Multiphysics. It avoids a perfectly discontinuous time switch, which can make the transient solver work harder. If your version does not accept `flc2hs`, replace `gtime` with `1` for a constant source during initial testing.

8 Assign silicon material properties

8.1 Create a blank material

1. In Model Builder, right-click **Materials**.
2. Choose **Blank Material**.
3. Rename the material **Silicon_reduced_constant_properties**.
4. In the Settings window, under **Material Contents**, add the following properties manually if they are not already listed:

- **Thermal conductivity**
- **Density**
- **Heat capacity at constant pressure**

8.2 Enter material values

In the material property table, enter:

Property	Expression to type	Meaning
Thermal conductivity	kSi	Fourier heat conduction coefficient
Density	rhoSi	mass per volume
Heat capacity at constant pressure	CpSi	energy per mass per kelvin

Physics meaning. These constant properties are sufficient for the first verification model. Temperature-dependent silicon properties can be added later, but a constant-property model is easier to verify against analytic diffusion and energy-balance tests.

9 Set up Heat Transfer in Solids

9.1 Check the domain equation

1. Click **Heat Transfer in Solids (ht)** in the Model Builder.
2. Click the default domain feature, usually called **Solid 1**.
3. Confirm that the selected domain is the rectangle.
4. Under **Heat Conduction**, confirm that thermal conductivity is taken from the material.
5. Under **Thermodynamics**, confirm that density and heat capacity are taken from the material.

9.2 Initial temperature

1. Under **Heat Transfer in Solids (ht)**, click **Initial Values 1**.
2. In the field **Temperature**, type T0.

This sets

$$T(x, y, 0) = T_0. \quad (9.1)$$

9.3 Add a volumetric heat source

1. Right-click **Heat Transfer in Solids (ht)**.
2. Choose **Domain** → **Heat Source**.
3. Rename it **Radiation and ballistic heat source**.
4. In the Settings window, make sure the rectangular domain is selected.
5. In the field **Heat source**, type Q_{tot} .

This implements the right-hand side of Eq. (1.2).

10 Boundary conditions

10.1 Default thermal insulation

In the Heat Transfer in Solids interface, exterior boundaries are usually thermally insulated by default unless another boundary condition is added. The mathematical condition is

$$-\mathbf{n} \cdot k \nabla T = 0. \quad (10.1)$$

This means no heat leaves the computational domain through that boundary.

10.2 Verify the default insulation

1. Expand **Heat Transfer in Solids (ht)**.
2. Look for **Thermal Insulation 1**. If it exists, click it.
3. Confirm that all exterior boundaries are selected.

10.3 Add fixed-temperature contacts, optional

Use this only if you want a heat-sink boundary rather than a thermally isolated device.

1. Right-click **Heat Transfer in Solids (ht)**.
2. Choose **Boundary** → **Temperature**.

3. Rename it **Left heat sink**.
4. Select the left vertical boundary of the rectangle.
5. In **Temperature**, type T_{left} .
6. Right-click **Heat Transfer in Solids (ht)** again.
7. Choose **Boundary** → **Temperature**.
8. Rename it **Right heat sink**.
9. Select the right vertical boundary.
10. In **Temperature**, type T_{right} .

Interpretation. Thermal insulation is useful for energy-conservation verification. Fixed-temperature boundaries are useful for modeling ideal contacts or heat sinks. Do not mix them accidentally: a fixed-temperature boundary can remove energy from the domain and will change the total thermal inventory.

11 Mesh creation

11.1 Create a controlled triangular mesh

1. In Model Builder, click **Mesh 1**.
2. In the Settings window, set **Sequence type** to **User-controlled mesh** if available.
3. Right-click **Mesh 1**.
4. Choose **Free Triangular**.
5. Right-click **Mesh 1**.
6. Choose **Size**.
7. In **Maximum element size**, type h_{max} .
8. In **Minimum element size**, type $h_{\text{max}}/5$.
9. In **Maximum element growth rate**, type 1.25.
10. Click **Build All**.

11.2 Mesh physics explanation

The temperature source has characteristic widths σ_{dep} and σ_{ball} . The mesh must resolve the smaller one. A practical initial rule is

$$h_{\text{max}} \lesssim \frac{\sigma_{\text{dep}}}{2} \quad (11.1)$$

near the hotspot. For a more reliable model, use local mesh refinement near the source center.

11.3 Optional local refinement near the hotspot

1. Right-click **Geometry 1**.
2. Choose **Circle**.
3. Set **Radius** to $3 \cdot \text{sigDep}$.
4. Set center **x** to **x0** and **y** to **y0**.
5. Build the geometry. If asked about forming a union, keep interior boundaries if you want a refinement region.
6. In **Mesh 1**, add a **Size** node assigned to the circular region or nearby boundaries.
7. Use maximum element size $\text{sigDep}/4$ in the source region.

12 Set the time-dependent study

12.1 Open Study 1

1. In Model Builder, expand **Study 1**.
2. Click **Step 1: Time Dependent**.
3. In the field **Times**, type

```
range(0,dtout,tend)
```

This asks COMSOL Multiphysics to output the solution from $t = 0$ to $t = t_{\text{end}}$ every Δt_{out} . The solver may use smaller internal time steps.

12.2 Solver tolerance

1. Click **Study 1**.

2. In the Settings window, find **Study Settings** or **Solver Configurations**.
3. For a first run, keep the default time-dependent solver.
4. For a verification run, set **Relative tolerance** to $1e-4$ or $1e-5$.

12.3 Compute

1. Click **Study 1**.
2. Click **Compute**.
3. Wait for the progress window to finish.

If the solver fails, first reduce Q_0 , increase `tpulse`, or use `gtime=1` to remove the smoothed time switch as a debugging step.

13 Postprocessing and plots

13.1 Temperature surface plot

1. After the solve finishes, expand **Results**.
2. A default **Temperature** plot group may already exist.
3. If not, right-click **Results** and choose **2D Plot Group**.
4. Rename it **Temperature map**.
5. Right-click the plot group and choose **Surface**.
6. In **Expression**, type `T`.
7. In **Unit**, type `K`.
8. Click **Plot**.

13.2 Temperature rise plot

For radiation effects, the temperature rise is often more informative than absolute temperature.

1. Right-click **Results** → **2D Plot Group**.
2. Rename it **Temperature rise**.

3. Right-click it and choose **Surface**.
4. In **Expression**, type $T-T_0$.
5. In **Unit**, type K.
6. Click **Plot**.

13.3 Line plot through the hotspot

1. Right-click **Results**.
2. Choose **1D Plot Group**.
3. Rename it **Centerline temperature**.
4. Right-click it and choose **Line Graph**.
5. For the dataset, choose the solution dataset.
6. Set the line from $(0, Ly/2)$ to $(Lx, Ly/2)$ if the GUI provides coordinate fields. Otherwise create a cut line dataset under **Results** $\rightarrow$ **Datasets** $\rightarrow$ **Cut Line 2D**.
7. In **Expression**, type $T-T_0$.
8. Click **Plot**.

14 Derived values and energy checks

14.1 Create an integration operator

1. In Model Builder, right-click **Component 1** $\rightarrow$ **Definitions**.
2. Choose **Component Couplings** $\rightarrow$ **Integration**.
3. Rename it **intop_domain**.
4. Select the silicon rectangle domain.

14.2 Total excess thermal energy

The excess thermal energy per out-of-plane thickness is

$$E_{\text{th}}(t) = \int_{\Omega} \rho_m c_p (T - T_0) d\Omega. \quad (14.1)$$

In COMSOL Multiphysics, evaluate it as:

```
intop_domain(rhoSi*CpSi*(T-T0))
```

14.3 Evaluate global quantities

1. Right-click **Results** → **Derived Values**.
2. Choose **Global Evaluation**.
3. In the expression table, type `intop_domain(rhoSi*CpSi*(T-T0))`.
4. Add a second expression `intop_domain(Qtot)`.
5. Click **Evaluate**.

The first expression gives total excess thermal energy per out-of-plane thickness. The second gives total input power per out-of-plane thickness.

15 Verification tutorial 1: uniform equilibrium

15.1 Purpose

This test checks whether the solver preserves a constant temperature when there is no source and no heat loss.

15.2 Steps

1. Go to **Component 1** → **Definitions** → **Variables**.
2. Temporarily change `Qtot` to `0[W/m^3]`.
3. Keep all exterior boundaries thermally insulated.
4. Keep **Initial Values 1** as `T0`.
5. Click **Study 1** → **Compute**.
6. Plot `T-T0`.

15.3 Expected result

The temperature rise should be zero everywhere:

$$T(x, y, t) - T_0 = 0. \quad (15.1)$$

If the model produces a nonzero temperature change, there is likely an unintended source, boundary condition, or initial-value mismatch.

16 Verification tutorial 2: uniform source in insulated domain

16.1 Purpose

This test checks the heat-capacity term. If the source is spatially uniform and boundaries are insulated, the temperature should rise uniformly as

$$T(t) = T_0 + \frac{Q_{\text{uniform}}}{\rho_m c_p} t. \quad (16.1)$$

16.2 Steps

1. In **Parameters**, add `Quni = 1e12[W/m^3]`.
2. In **Variables**, set `Qtot` to `Quni`.
3. Keep all boundaries thermally insulated.
4. Set `tend` to `100[ns]`.
5. Compute.
6. Use **Derived Values** → **Global Evaluation** to evaluate `maxop1(T)` if you have a maximum operator, or plot `T` and inspect uniformity.

16.3 Expected result

The temperature should remain spatially uniform. At time t , compare with

$$\Delta T_{\text{analytic}} = \frac{Q_{\text{uni}} t}{\rho_m c_p}. \quad (16.2)$$

This test is a direct check of the transient mass matrix and volumetric source term.

17 Verification tutorial 3: Gaussian diffusion

17.1 Purpose

This test checks the diffusion operator. Turn off the source after setting a Gaussian initial temperature field. The hotspot should spread and decrease in amplitude.

17.2 Create a Gaussian initial condition

1. In **Parameters**, add $\text{dT0} = 10[\text{K}]$.
2. Add $\text{sigT0} = 1[\mu\text{m}]$.
3. In **Component 1** → **Definitions** → **Variables**, add

$$\text{Tgauss} = \text{T0} + \text{dT0} \cdot \exp(-((x-x0)^2 + (y-y0)^2) / (2 \cdot \text{sigT0}^2))$$

4. Click **Heat Transfer in Solids** → **Initial Values 1**.
5. Set **Temperature** to **Tgauss**.
6. Set **Qtot** to $0[\text{W}/\text{m}^3]$.
7. Keep boundaries insulated, or make the domain much larger than the hotspot for early-time comparison.
8. Compute.

17.3 Expected result

The hotspot spreads with time. In an effectively infinite domain, the width grows like

$$\sigma^2(t) = \sigma_0^2 + 4\alpha t. \quad (17.1)$$

The total excess energy

$$\int_{\Omega} \rho_m c_p (T - T_0) d\Omega \quad (17.2)$$

should remain approximately constant if boundaries are insulated.

18 Verification tutorial 4: fixed-temperature boundaries

18.1 Purpose

This test checks boundary-condition enforcement. With no source and left/right boundaries fixed at different temperatures, the steady solution should become nearly linear in x .

18.2 Steps

1. Set Q_{tot} to 0 [W/m³].
2. Set $T_{\text{left}} = 310$ [K] and $T_{\text{right}} = 300$ [K].
3. Add left and right **Temperature** boundary conditions as described earlier.
4. Use a sufficiently long **tend**, for example 10 [us] or longer depending on geometry.
5. Compute.
6. Plot a centerline T profile from left to right.

18.3 Expected result

The temperature should approach

$$T(x) \approx T_{\text{left}} + \frac{T_{\text{right}} - T_{\text{left}}}{L_x} x. \quad (18.1)$$

This test is useful because it checks that the thermal conductivity matrix and Dirichlet boundary conditions are behaving properly.

19 Route B: Coefficient Form PDE implementation

19.1 When to use this route

Use this route if you do not have the Heat Transfer Module or if you want a closer one-to-one connection to the MATLAB FEM equation. In this case the dependent variable is still temperature T , but you enter the heat equation as a generic PDE.

19.2 Create the PDE model

1. Start a new COMSOL Multiphysics model.
2. In **Model Wizard**, choose **2D**.
3. In **Select Physics**, expand **Mathematics**.

4. Expand **PDE Interfaces**.
5. Select **Coefficient Form PDE**.
6. Click **Add**.
7. In the dependent variable field, change the dependent variable name to **T** if possible.
8. Choose **Time Dependent** study.
9. Click **Done**.

19.3 Coefficient form equation

The coefficient form PDE is commonly written in COMSOL Multiphysics as

$$e_a \frac{\partial^2 u}{\partial t^2} + d_a \frac{\partial u}{\partial t} + \nabla \cdot (-c \nabla u - \alpha u + \gamma) + \beta \cdot \nabla u + a u = f. \quad (19.1)$$

For heat diffusion with $u = T$, use

$$d_a = \rho_m c_p, \quad c = k, \quad f = Q_{\text{tot}}, \quad (19.2)$$

and set the remaining coefficients to zero.

19.4 Enter PDE coefficients

Click the main **Coefficient Form PDE** domain feature and enter:

Coefficient	Expression	Meaning
ea	0	no second time derivative
da	rhoSi*CpSi	heat-capacity coefficient
c	kSi	thermal conductivity
a	0	no linear reaction/sink in base model
f	Qtot	volumetric heat source
alpha	0	no conservative convection-like flux
beta	0	no nonconservative advection
gamma	0	no prescribed internal flux

19.5 Boundary condition in PDE route

For zero-flux boundaries, use the default no-flux condition if present. If you need to specify it manually, set the normal flux to zero:

$$\mathbf{n} \cdot (-c \nabla T) = 0. \quad (19.3)$$

For fixed-temperature boundaries, add a Dirichlet boundary condition and set $T = T_b$.

20 Adding imported radiation-deposition maps

20.1 Why import maps

The analytic Gaussian source is useful for verification, but the project ultimately needs source maps from radiation transport or other preprocessing. A map may contain deposited energy density, power density, or event-normalized heat deposition.

20.2 Prepare a text file

Use a plain text or CSV file with columns such as

```
x,y,Q
0.0e-6,0.0e-6,1.2e15
0.1e-6,0.0e-6,1.1e15
...
```

Use SI units if possible.

20.3 Import interpolation function

1. Right-click **Global Definitions**.
2. Choose **Functions** → **Interpolation**.
3. Rename the function **Qmap**.
4. In **Source**, choose **File**.
5. Browse to the CSV or text file.
6. Set the number of arguments to 2.
7. Set argument units to **m** and **m**.

8. Set function unit to W/m³ if the map is a power density.
9. Click **Import** or **Plot** to verify the function.

20.4 Use imported map in source term

In **Variables**, set

```
Qdep = Qmap(x,y)*gtime
Qtot = Qdep + Qball
```

If the map already includes time-integrated energy density E_{dep} in J/m³, convert it to power density by dividing by a thermalization time:

```
Qdep = Edep_map(x,y)/tpulse*gtime
```

21 Adding defect-related thermal coupling

21.1 Temperature-dependent defect kinetics

Even if Module 4 only solves temperature, its output is used by Module 3. Defect diffusion and annealing often use Arrhenius forms:

$$D_i(T) = D_{i0} \exp\left(-\frac{E_{a,i}}{k_B T}\right), \quad (21.1)$$

$$k_i(T) = \nu_i \exp\left(-\frac{E_{m,i}}{k_B T}\right). \quad (21.2)$$

The COMSOL Multiphysics temperature map can be exported to MATLAB so that Module 3 can use these temperature-dependent coefficients.

21.2 Defect reaction heat source

If a defect reaction releases or absorbs heat, include

$$Q_{\text{defect}} = \sum_r \Delta H_r R_r, \quad (21.3)$$

where ΔH_r is reaction enthalpy per reaction and R_r is reaction rate per volume. In the first tutorial model set $Q_{\text{defect}} = 0$. Add it only after the base heat equation is verified.

22 Exporting data to MATLAB

22.1 Export a temperature field

1. Right-click **Results**.
2. Choose **Export** → **Data**.
3. In the Settings window, set **Dataset** to the time-dependent solution.
4. In **Expression**, type T or T-T0.
5. Choose the desired time step under **Time selection**.
6. Set **Filename** to something like module4_T_field.csv.
7. Click **Export**.

22.2 Export centerline data

1. Create a **Cut Line 2D** dataset from $(0, Ly/2)$ to $(Lx, Ly/2)$.
2. Right-click **Results** → **Export** → **Data**.
3. Select the cut-line dataset.
4. Export **x**, **T**, and **T-T0**.

23 Mesh convergence workflow

23.1 Why convergence matters

The FEM solution should not depend strongly on arbitrary mesh size. If the temperature peak changes significantly when the mesh is refined, the mesh is too coarse.

23.2 Manual mesh convergence steps

1. Run the model with **hmax** = 0.8[um].
2. Record peak T-T0, total excess energy, and centerline temperature.
3. Change **hmax** = 0.4[um].
4. Rebuild the mesh and recompute.

5. Change `hmax = 0.2[um]`.
6. Rebuild and recompute again.
7. Compare the results.

A reasonable convergence indicator is

$$\epsilon_h = \frac{|T_{\max}^{(h)} - T_{\max}^{(h/2)}|}{|T_{\max}^{(h/2)} - T_0|}. \quad (23.1)$$

For a demonstration model, $\epsilon_h < 5\%$ may be acceptable. For a quantitative model, use a stricter threshold.

24 Time-step sensitivity workflow

24.1 Why time-step checks matter

Even though COMSOL Multiphysics uses adaptive internal time steps, the output interval and solver tolerances can affect apparent pulse dynamics. A short pulse requires time resolution fine enough to capture its energy input.

24.2 Steps

1. Run with `dtout = tpulse`.
2. Run with `dtout = tpulse/2`.
3. Run with `dtout = tpulse/5`.
4. Compare peak temperature and integrated input energy.
5. If the peak changes strongly, refine the time output and solver tolerances.

25 Common mistakes and fixes

Symptom	Likely cause	Fix
Temperature does not change	<code>Qtot</code> is zero or not assigned to domain	Check Heat Source domain selection and expression

Temperature rise is enormous	Q_0 too large or units wrong	Verify W/m^3 vs J/m^3
Hotspot is blocky	mesh too coarse near source	reduce <code>hmax</code> or add local refinement
Energy is not conserved in insulated test	hidden temperature boundary or flux condition	remove fixed-temperature boundaries
Solver fails at first time step	source turns on too sharply	use smoothed <code>gtime</code> , reduce Q_0 , or increase <code>tpulse</code>
Imported map gives wrong scale	map units inconsistent	convert deposited energy to power density carefully
COMSOL expression turns red	syntax or unit mismatch	check brackets, multiplication signs, and units

26 Mapping COMSOL objects to MATLAB FEM objects

Physics concept	COMSOL object	MATLAB FEM analogue
Domain Ω	Rectangle geometry	node coordinates and element connectivity
Shape functions N_a	selected FEM element order	basis functions inside element routines
Mass matrix M	density and heat capacity in transient heat equation	assembled heat-capacity matrix
Conductivity matrix K	thermal conductivity in Heat Transfer interface	assembled diffusion matrix
Source vector F	Heat Source <code>Qtot</code>	assembled load vector
Initial condition	Initial Values node	initial nodal temperature vector
Boundary conditions	Temperature or Thermal Insulation nodes	Dirichlet or Neumann boundary handling
Solution vector	Temperature field <code>T</code>	nodal temperature vector
Postprocessing	Surface/line/global plots	MATLAB plotting and summary functions

27 Recommended model-building order

For reliable work, build the model in this order:

1. Constant material properties.
2. No source, insulated boundaries: verify uniform equilibrium.
3. Uniform source, insulated boundaries: verify linear temperature rise.
4. Gaussian initial condition, no source: verify diffusion and energy conservation.
5. Gaussian source pulse: verify source-driven heating.
6. Add broadened ballistic source term.
7. Add imported radiation-deposition maps.
8. Add thermal boundary conditions or heat sinks.
9. Export data to MATLAB and compare.

Do not begin with the fully coupled project model. Build from verification cases upward.

28 Final trust checklist

Before trusting a Module 4 COMSOL Multiphysics result, check the following:

- The material properties use consistent SI units.
- The heat source has units of W/m^3 , not J/m^3 , unless divided by a time scale.
- The source region is resolved by the mesh.
- The chosen boundary conditions match the physical device assumption.
- Uniform equilibrium test passes.
- Uniform source test gives $dT/dt = Q/(\rho_m c_p)$.
- Gaussian diffusion test conserves energy under insulated boundaries.
- Mesh refinement changes peak temperature by an acceptably small amount.
- Time-step sensitivity has been checked for pulsed sources.
- Exported temperature maps use the same coordinate system expected by MATLAB.

29 Detailed COMSOL node tree expected after setup

After completing Route A, the Model Builder tree should have the following logical structure. The exact labels can vary slightly by COMSOL Multiphysics version, but the physical content should match.

```

Global Definitions
  Parameters 1
Component 1
  Definitions
    Thermal source variables
    intop_domain
  Geometry 1
    Rectangle 1
  Materials
    Silicon_reduced_constant_properties
  Heat Transfer in Solids (ht)
    Solid 1
    Initial Values 1
    Radiation and ballistic heat source
    Thermal Insulation 1
    Optional: Left heat sink
    Optional: Right heat sink
  Mesh 1
    Free Triangular 1
    Size 1
Study 1
  Step 1: Time Dependent
Results
  Temperature map
  Temperature rise
  Centerline temperature
  Derived Values

```

If your tree has extra default nodes, that is acceptable. The critical issue is that the rectangular domain must have one material, one transient heat equation, one initial condition, and the intended boundary conditions.

30 Detailed expression-entry rules in COMSOL

30.1 Multiplication signs

Always type multiplication explicitly. Use `rhoSi*CpSi`, not `rhoSi CpSi`. COMSOL Multiphysics usually marks invalid expressions in red, but dimensional mistakes can still survive if the units accidentally reduce to something plausible.

30.2 Units inside brackets

Use square brackets for units:

```
20[um]
1e15[W/m^3]
300[K]
700[J/(kg*K)]
```

This tells COMSOL Multiphysics both the numerical value and the physical dimension. The software internally converts to SI units when solving.

30.3 Coordinate variables

In a 2D model the spatial coordinates are normally `x` and `y`. Expressions such as

```
(x-x0)^2+(y-y0)^2
```

are evaluated pointwise at quadrature points during FEM assembly. This is exactly how the source vector is generated from a spatial heat source.

30.4 Time variable

In a time-dependent study, COMSOL Multiphysics uses `t` for time. Therefore the expression

```
gtime = flc2hs(tpulse-t,0.1*tpulse)
```

is time dependent. If the study is stationary, `t` is not defined unless you introduce it as a parameter.

31 More detailed boundary-condition options

31.1 Thermal insulation

Thermal insulation means

$$-\mathbf{n} \cdot k \nabla T = 0. \quad (31.1)$$

Use it when the simulation is meant to represent an isolated local volume or when you are running an energy-conservation verification test.

Click-by-click:

1. Right-click **Heat Transfer in Solids (ht)**.
2. Choose **Boundary** → **Thermal Insulation** if it is not already present.
3. Select the outer boundaries that should have zero normal heat flux.
4. Click **Build Selected** is not needed for physics; just ensure the selection is active.

31.2 Fixed temperature

Fixed temperature means

$$T = T_b. \quad (31.2)$$

Use it for idealized contacts or heat sinks. This is a Dirichlet thermal boundary condition.

Click-by-click:

1. Right-click **Heat Transfer in Solids (ht)**.
2. Choose **Boundary** → **Temperature**.
3. Select the boundary connected to an ideal reservoir.
4. Type T_{left} , T_{right} , or another parameter in the **Temperature** field.

31.3 Prescribed heat flux

A heat flux boundary applies

$$-\mathbf{n} \cdot k \nabla T = q_0'' \quad (31.3)$$

This is useful for a laser-like or surface-heating approximation.

Click-by-click:

1. Right-click **Heat Transfer in Solids (ht)**.
2. Choose **Boundary** → **Heat Flux**.
3. Select the boundary receiving surface heat.
4. In the inward heat flux field, type an expression such as `qSurf0*gtime`.
5. Add `qSurf0 = 1e6[W/m^2]` in **Parameters** if needed.

31.4 Convective cooling

A convective boundary approximates heat loss to an environment:

$$-\mathbf{n} \cdot k \nabla T = h(T_{\infty} - T) \quad (31.4)$$

with heat-transfer coefficient h . This is not usually the first choice for a silicon microdevice cross-section unless the boundary really represents an interface to a surrounding medium.

Click-by-click:

1. Right-click **Heat Transfer in Solids (ht)**.
2. Choose **Boundary** → **Heat Flux**.
3. In the Settings window, select the convective heat flux option if available.
4. Type **hconv** for the heat transfer coefficient.
5. Type **Text** for the external temperature.
6. Add parameters **hconv** = 10[W/(m²*K)] and **Text** = 300[K].

32 How to create selections for boundaries

Selections make the model less fragile. Rather than repeatedly clicking boundaries manually, define named selections.

32.1 Left boundary selection

1. Right-click **Component 1** → **Definitions**.
2. Choose **Selections** → **Explicit**.
3. Rename it **left_boundary**.
4. Change the geometric entity level to **Boundary**.
5. Click the left edge of the rectangle in the Graphics window.
6. Click the **+** button to add it to the selection.

32.2 Right boundary selection

Repeat the same steps and rename the selection **right_boundary**. Select the right edge of the rectangle.

32.3 Source-region selection, optional

If you created a circle or subdomain around the hotspot, create a selection named `source_region`. This is useful when you want a different mesh size near the deposition center.

33 How to inspect the mesh

33.1 Plot mesh quality

1. Right-click **Results**.
2. Choose **More Plot Groups** → **Mesh Plot** if available.
3. Select **Mesh 1**.
4. Click **Plot**.

33.2 Check element size near the source

Zoom into the source center (x_0, y_0) . The Gaussian deposition region should contain several elements across its width. If it is represented by only one or two elements, the peak temperature will be inaccurate.

33.3 Check mesh statistics

1. Click **Mesh 1**.
2. Look for **Statistics** in the Settings or toolbar.
3. Record the number of triangular elements and minimum element quality.

Low-quality elements can cause inaccurate gradients or solver difficulty. For this rectangular model, the mesh quality should generally be good.

34 How to duplicate studies for verification cases

Do not overwrite the main radiation-heating study each time you run a verification case. Instead, duplicate studies.

1. Right-click **Study 1**.

2. Choose **Duplicate**.
3. Rename the duplicate `Study_uniform_source_verification` or another descriptive name.
4. Change only the parameters or variables needed for that verification case.
5. Compute the duplicate study.

This preserves the original setup and makes it easier to compare cases.

35 Parametric sweep for mesh refinement

35.1 Add the sweep

1. Right-click **Study 1**.
2. Choose **Parametric Sweep**.
3. In the parameter table, select `hmax`.
4. Enter values such as `0.8[um]` `0.4[um]` `0.2[um]`.
5. Make sure the mesh uses `hmax` as its maximum element size.
6. Click **Compute**.

35.2 What to record

For each mesh value, record:

- maximum temperature rise, $\max(T - T_0)$,
- total excess energy, $\int \rho_m c_p (T - T_0) d\Omega$,
- centerline profile through the hotspot,
- number of mesh elements.

36 Parametric sweep for ballistic source width

The parameter `sigBall` controls the spatial width of the broadened ballistic source. To understand sensitivity:

1. Right-click **Study 1**.

2. Choose **Parametric Sweep**.
3. Select parameter `sigBall`.
4. Type values 1[um] 2[um] 4[um] 8[um].
5. Compute.
6. Compare peak temperature rise and spatial spread.

Increasing `sigBall` should generally reduce the peak temperature and spread the heating over a larger region, if total energy is normalized consistently. If the total energy changes unintentionally, normalize the Gaussian source.

37 Source normalization issue

The Gaussian source expressions above are convenient but not automatically normalized to fixed total power. If you change `sigDep` or `sigBall`, the integral of the source also changes. For sensitivity studies where total energy must remain fixed, normalize the Gaussian.

For an infinite 2D Gaussian,

$$\int_{\mathbb{R}^2} \exp\left(-\frac{r^2}{2\sigma^2}\right) dA = 2\pi\sigma^2. \quad (37.1)$$

A normalized source can therefore use

$$Q_{\text{dep}} = \frac{P_{\text{dep}}}{2\pi\sigma_{\text{dep}}^2} f_{\text{dep}} g(t), \quad (37.2)$$

where P_{dep} is power per out-of-plane thickness. In a finite domain the exact integral differs slightly, so use `intop_domain(fdep)` for numerical normalization if needed.

In COMSOL Multiphysics, define

```
Qdep = Pdep*fdep/intop_domain(fdep)*gtime
```

where `Pdep` has units of W/m in a 2D model interpreted per unit out-of-plane thickness.

38 Interpreting 2D units carefully

A 2D COMSOL Multiphysics model represents a cross-section. Volumetric heat source terms still use W/m^3 , but integrated domain quantities are often interpreted per unit out-of-plane thickness. For example,

$$\int_{\Omega} Q dA \quad (38.1)$$

has units W/m, not W, because the missing third dimension has not been integrated.

If the physical device has an out-of-plane thickness L_z , the total power is approximately

$$P_{3D} = L_z \int_{\Omega} Q \, dA. \quad (38.2)$$

Add a parameter **Lz** if you want to convert 2D per-thickness quantities to total 3D values.

39 Adding temperature-dependent thermal conductivity

After the constant-property model is verified, one may use temperature-dependent conductivity. A simple linearized model is

$$k(T) = k_0 [1 + a_k(T - T_0)]. \quad (39.1)$$

In COMSOL Multiphysics:

1. Add parameter **ak** = -1e-3[1/K] or another chosen coefficient.
2. Add variable **kT** = **kSi***(1+**ak***(**T**-**T0**)).
3. In the material property for thermal conductivity, replace **kSi** with **kT**.

Do not introduce this before the constant-property verification tests pass. Temperature-dependent material properties make the PDE nonlinear and can hide simpler setup mistakes.

40 Adding a heat sink layer, optional model variant

If you want to model silicon attached to a thermal sink, add a second rectangle below the silicon.

1. In **Geometry 1**, add a second rectangle.
2. Set its width to **Lx** and height to **Lsink**.
3. Position it below the silicon, for example **y=-Lsink**.
4. Add parameter **Lsink** = 5[um].
5. Assign a separate material to the sink with its own k , ρ , and c_p .
6. Keep continuity at the internal boundary by default.
7. Apply fixed temperature or convection at the bottom of the sink.

This variant is closer to a real device mounted on a substrate or thermal path, but it is less useful for the simplest energy-conservation tests.

41 Creating probes for time traces

41.1 Point probe at hotspot center

1. Right-click **Definitions** under **Component 1**.
2. Choose **Probes** → **Point Probe** if available.
3. Set the point coordinates to x_0, y_0 .
4. Set expression to $T-T_0$.
5. Compute the study again.

A point probe gives the hotspot temperature rise as a function of time. This is often the clearest way to compare different source widths or thermal boundary conditions.

41.2 Domain probe for total thermal energy

If your COMSOL Multiphysics version supports domain probes:

1. Create a domain probe.
2. Set expression to $\rho_{\text{Si}} \cdot c_{\text{pSi}} \cdot (T - T_0)$.
3. Use integration over the silicon domain.

This gives a time trace of total thermal energy per out-of-plane thickness.

42 Report generation inside COMSOL, optional

1. Click **File** → **Report** or use the **Report** node if available.
2. Select model contents, parameters, physics settings, mesh, solver, and results.
3. Generate a PDF or HTML report for archival use.

This is not a substitute for the LaTeX project notes, but it is useful for recording exact COMSOL Multiphysics GUI settings.

43 How to compare with MATLAB Module 4

43.1 Use the same inputs

The MATLAB and COMSOL Multiphysics comparisons are meaningful only if the following match:

- domain size L_x, L_y ,
- material properties ρ_m, c_p, k ,
- initial condition,
- heat source amplitude and units,
- boundary conditions,
- output times.

43.2 Compare robust quantities first

Do not compare individual nodal values first because MATLAB and COMSOL Multiphysics may use different meshes. Start with:

- total excess thermal energy,
- maximum temperature rise,
- centerline profile interpolated onto common coordinates,
- qualitative hotspot spreading rate.

43.3 Then compare field maps

Export the COMSOL Multiphysics temperature field to CSV and interpolate it onto the MATLAB grid or mesh. Compare

$$\Delta T_{\text{err}} = T_{\text{COMSOL}} - T_{\text{MATLAB}}. \quad (43.1)$$

A useful normalized error is

$$\epsilon_T = \frac{\|T_{\text{COMSOL}} - T_{\text{MATLAB}}\|_2}{\|T_{\text{COMSOL}} - T_0\|_2}. \quad (43.2)$$

44 Beginner navigation summary

For quick reference, the main click paths are:

Task	COMSOL path
Create model	Model Wizard → 2D → Heat Transfer in Solids → Time Dependent
Parameters	Global Definitions → Parameters 1
Geometry	Component 1 → Geometry 1 → Rectangle
Variables	Component 1 → Definitions → Variables
Material	Materials → Blank Material
Heat source	Heat Transfer in Solids → right-click → Domain → Heat Source
Fixed temperature	Heat Transfer in Solids → right-click → Boundary → Temperature
Insulation	Heat Transfer in Solids → Thermal Insulation
Mesh	Mesh 1 → Free Triangular and Size
Time range	Study 1 → Step 1: Time Dependent
Plot	Results → 2D Plot Group → Surface
Export	Results → Export → Data

45 Detailed verification worksheets

This section is intentionally repetitive. It is written as a beginner worksheet so that a user can reproduce the verification sequence without guessing which setting to change.

45.1 Worksheet A: baseline radiation pulse case

1. Open **Global Definitions → Parameters 1**.
2. Confirm $Q_0 = 1e15 [W/m^3]$.
3. Confirm $\eta_{Ball} = 0.3$.
4. Confirm $t_{pulse} = 10 [ns]$.
5. Open **Component 1 → Definitions → Thermal source variables**.
6. Confirm $Q_{dep} = Q_0 * f_{dep} * g_{time}$.
7. Confirm $Q_{ball} = \eta_{Ball} * Q_0 * f_{ball} * g_{time}$.

8. Confirm $Q_{\text{tot}} = Q_{\text{dep}} + Q_{\text{ball}}$.
9. Open **Heat Transfer in Solids** → **Initial Values 1**.
10. Confirm T_0 is used as the initial temperature.
11. Open the heat source node.
12. Confirm the source expression is Q_{tot} .
13. Compute the time-dependent study.
14. Plot $T - T_0$.
15. Evaluate `intop_domain(rhoSi*CpSi*(T-T0))` at several times.

Expected behavior: temperature rises near the source during the pulse and then spreads by diffusion. The broadened ballistic source should make the temperature map wider than the narrow deposition-only source.

45.2 Worksheet B: local deposition only

1. Go to **Parameters 1**.
2. Set `etaBall = 0`.
3. Recompute.
4. Plot $T - T_0$ at the same time as the baseline case.
5. Compare peak temperature rise against the baseline.

Expected behavior: the source is narrower and the peak temperature is usually higher, assuming total deposited power is not normalized separately.

45.3 Worksheet C: ballistic broadening sensitivity

1. Set `etaBall = 0.3`.
2. Set `sigBall = 1[um]` and compute.
3. Record peak $T - T_0$.
4. Set `sigBall = 2.5[um]` and compute.
5. Record peak $T - T_0$.

6. Set `sigBall = 5[um]` and compute.
7. Record peak $T-T_0$.

Expected behavior: broadening redistributes heat. If the source is not normalized to fixed total power, changing `sigBall` also changes total input power. Therefore, when doing a physically meaningful sensitivity study, use the normalized source form described earlier.

45.4 Worksheet D: insulated energy accounting

1. Remove or disable all fixed-temperature boundaries.
2. Ensure only thermal insulation is active on the outer boundary.
3. Use a finite pulse source.
4. Compute.
5. Evaluate `intop_domain(Qtot)` as a function of time.
6. Evaluate `intop_domain(rhoSi*CpSi*(T-T0))` as a function of time.
7. The time integral of input power should match the increase in thermal energy, within numerical tolerance.

This is one of the most important checks. If energy is disappearing in an insulated model, the boundary conditions or source units are probably wrong.

46 Detailed solver settings for difficult transient runs

46.1 Open the solver configuration

1. Expand **Study 1**.
2. Expand **Solver Configurations** if it is visible.
3. Click **Time-Dependent Solver**.
4. If solver nodes are hidden, right-click **Study 1** and choose **Show Default Solver** or **Get Initial Value/Solver Configuration**, depending on version.

46.2 Suggested time stepping

For smooth sources, the default BDF time stepping is usually fine. For short pulses, constrain the maximum time step.

1. Click **Time-Dependent Solver**.
2. Find the **Time Stepping** section.
3. Set **Steps taken by solver** to **Free** for normal use.
4. If the pulse is missed, set **Maximum step** to $t_{\text{pulse}}/10$.
5. Keep **Relative tolerance** at $1e-4$ initially.
6. Tighten to $1e-5$ for final verification.

46.3 Linear solver choice

For this scalar thermal problem, the default direct solver is usually robust. For very large meshes, iterative solvers may be faster but require more numerical care. For beginner use, do not change the linear solver until the base model is working.

47 Detailed instructions for saving model variants

47.1 Use separate files for major variants

Use separate .mph files for major branches:

```
module4_thermal_2d_base.mph
module4_thermal_2d_uniform_source_verification.mph
module4_thermal_2d_gaussian_diffusion_verification.mph
module4_thermal_2d_imported_geant4_map.mph
module4_thermal_2d_heat_sink_boundary.mph
```

This prevents accidental loss of the verified base model.

47.2 Use descriptive study names

Inside one model, rename studies clearly:

```
Study 1 - Radiation pulse
Study 2 - Uniform equilibrium test
Study 3 - Uniform source test
```

Study 4 - Gaussian diffusion test
Study 5 - Mesh convergence

A descriptive tree is part of reproducibility.

48 Detailed imported-map workflow

48.1 Check coordinate convention before import

Before importing a radiation deposition map, confirm:

- Are coordinates in meters, micrometers, or centimeters?
- Is the origin at the lower-left corner, center, or another reference point?
- Does x correspond to lateral position and y to depth?
- Is the map a deposited energy density J/m^3 , deposited dose, or power density W/m^3 ?

48.2 Convert deposited energy to heat source

If the map gives energy density $E_{\text{dep}}(x, y)$ in J/m^3 , then a simple pulse heating model is

$$Q_{\text{dep}}(x, y, t) = \frac{E_{\text{dep}}(x, y)}{\tau_{\text{th}}} g(t), \quad (48.1)$$

where τ_{th} is an assumed thermalization time. In COMSOL Multiphysics type:

`Qdep = EdepMap(x,y)/tauTh*gtime`

with `tauTh` defined in Parameters.

48.3 Interpolation settings to check

When setting up the interpolation function:

1. Choose interpolation, not analytic function.
2. Set the argument dimension to two.
3. Set argument units correctly.
4. Set extrapolation to constant or nearest only if you understand what happens outside the data range.
5. Plot the imported function before using it in the heat equation.

49 Detailed postprocessing recipes

49.1 Plot temperature at selected times

1. Click the **Temperature rise** plot group.
2. In the Settings window, find **Data**.
3. Under **Time**, choose **Manual** or **From list** if available.
4. Select times such as 0, `tpulse`, `100[ns]`, and `1[us]`.
5. Click **Plot**.

49.2 Create an animation

1. Right-click **Results**.
2. Choose **Animation** → **Player** or **File** depending on version.
3. Select the temperature-rise plot group.
4. Choose all time steps.
5. Play the animation to verify qualitative heat spreading.

49.3 Export plot images

1. Right-click **Results** → **Export**.
2. Choose **Image**.
3. Select the desired plot group.
4. Choose PNG or PDF format.
5. Export with a descriptive filename.

50 Physical interpretation of common outcomes

50.1 High narrow peak

A high narrow temperature peak means the energy deposition is localized and diffusion has not yet spread the heat. Check whether the source width is physically reasonable and whether the mesh resolves it.

50.2 Broad low-amplitude hotspot

A broad low-amplitude hotspot can indicate ballistic spreading, a large source width, or strong thermal diffusion. Compare early and late times: ballistic broadening appears immediately through the source, while Fourier diffusion broadens progressively with time.

50.3 Boundary-dominated cooling

If fixed-temperature boundaries are close to the hotspot, the temperature may decay rapidly. This can be physical if the device is strongly heat-sunk, but it can also be an artifact of placing artificial boundaries too close.

50.4 Nearly uniform temperature rise

If the temperature becomes nearly uniform, either the source is uniform, diffusion has had enough time to smooth the field, or the mesh/plot range is hiding small gradients. Use centerline plots and numerical maxima to distinguish these cases.

51 What not to include in the first COMSOL model

For the first COMSOL Multiphysics implementation, do not include all possible physics at once. Specifically, postpone:

- temperature-dependent material properties,
- nonlinear defect reaction heat,
- carrier Joule heating,
- full electron-phonon nonequilibrium,
- anisotropic thermal conductivity,
- moving boundaries,
- stochastic event-by-event deposition.

Each of these can be added later, but each also adds a new failure mode. The beginner model should first prove that the heat equation, source term, mesh, boundaries, and time stepping are correct.

52 Standalone reproduction checklist

A person with no prior COMSOL Multiphysics experience should be able to reproduce the base model by following this short checklist after reading the detailed sections:

1. Open Model Wizard.
2. Select 2D.
3. Add Heat Transfer in Solids.
4. Add Time Dependent study.
5. Type all global parameters.
6. Build a rectangle with width L_x and height L_y .
7. Add source variables Q_{dep} , Q_{ball} , and Q_{tot} .
8. Add silicon material properties ρ_{Si} , Cp_{Si} , and k_{Si} .
9. Set initial temperature to T_0 .
10. Add a domain Heat Source equal to Q_{tot} .
11. Use thermal insulation on all exterior boundaries for the first run.
12. Build a triangular mesh with maximum size h_{max} .
13. Set output times to `range(0,dtout,tend)`.
14. Compute.
15. Plot $T-T_0$.
16. Evaluate total excess energy using `intop_domain(rhoSi*CpSi*(T-T0))`.
17. Run the uniform equilibrium and uniform source verification cases.

53 Summary

This note showed how to implement Module 4 in COMSOL Multiphysics at a beginner click-by-click level. The Heat Transfer in Solids route is the most natural way to solve the transient thermal problem. The Coefficient Form PDE route is useful when one wants a direct match to the MATLAB FEM form $M\dot{\mathbf{T}} + K\mathbf{T} = \mathbf{F}$. The most important modeling discipline is to verify simple cases before adding radiation source maps and coupled physics. Once verified, the COMSOL Multiphysics model provides an independent FEM implementation of the Module 4 thermal solve and a useful comparison against the MATLAB code.